Types of Operating System

Tasks :-

1.Create a table comparing at least three types of operating systems (such as Windows, Linux, and Android) based on their main features, target devices, and examples of apps you use on each.

Ans :-

Operating System	Main Features	Target Devices	Example Apps
Windows	• User-friendly GUI with a Start Menu and taskbar • Massive software library • Strong gaming support • Regular security and feature updates	• Desktop PCs • Laptops • Tablets (e.g., Surface Pro) • Workstations / Servers	• Microsoft Office • Adobe Photoshop **/** Premiere Pro • Steam **/** Xbox Game Pass • Chrome **/** Edge browsers
Linux	• Open-source and highly customizable • Strong security and privacy focus • Lightweight, runs well on older hardware	• Servers / Cloud infrastructure • Developer workstations • Embedded systems • Olderor budget laptops	• LibreOffice • GIMP • VS Code / IntelliJ • Firefox / Thunderbird
Android	• Touch-optimized interface with widgets • Deep integration with Google services • Multi-window and notification management	• Smartphones • Tablets • Smart TVs • Wearables	• WhatsApp / Telegram • YouTube / Spotify • Google Maps • TikTok / Instagram

2.Identify the operating system running on your phone and your laptop/desktop. Write down two unique features of each OS that affect your daily use of popular apps like Instagram or Zomato.

Ans :-

Device	Operating System	Two Unique Features
Phone	Android	1. Touchscreen makes apps like Instagram and Zomato easy to use. 2. Notifications provide instant updates from apps.
Laptop/Desktop	Windows	1. Multitasking allows multiple apps and browser tabs to run together. 2. Large screen and keyboard make browsing Instagram and using Zomato more convenient.

3.List three security risks that are specific to mobile operating systems like Android or iOS, and explain how these risks could impact apps you use (for example, WhatsApp or Paytm).

Ans :-

Security Risk	Explanation	Impact on Apps like WhatsApp or Paytm
1. Malicious Apps & Malware	Users may install apps containing malware that can steal data or access device resources.	Malware could steal WhatsApp messages, contacts, or Paytm login/payment information.
2. Phishing & Fake Apps	Attackers may create fake versions of apps or send links that trick users into entering passwords, OTPs, or banking details.	A fake Paytm login page could steal account credentials or payment information.
3. Unpatched OS & App Vulnerabilities	If the mobile OS or apps are not updated, attackers may exploit known security weaknesses.	A vulnerability could allow unauthorized access to WhatsApp data or sensitive information used by Paytm.

4.Research and describe how a cloud-based operating system (such as Chrome OS or a virtual desktop environment) is different from a traditional desktop OS. Give one example of how this difference changes the way you access apps like YouTube or Flipkart.

Ans :-

- **Key Differences at a Glance :-**

Feature	Traditional OS (Windows/macOS)	Cloud-Based OS (Chrome OS / VDI)
Processing Location	Applications run locally on the device's CPU, RAM, and storage.	Applications run on remote servers in a data center or cloud; the local device only displays the interface.
Application Access	Software must be installed, updated, and managed individually on each device.	Apps are accessed via a web browser or streamed from the cloud; no local installation is required.
Performance Dependency	Tied to local hardware specifications; powerful hardware is needed for demanding tasks.	Depends heavily on internet bandwidth and latency; minimal local hardware requirements are sufficient.
Data Storage	Files and data are primarily stored on the local hard drive.	Files and data are stored on the cloud, accessible from any device.

Example: How This Changes Access to YouTube or Flipkart

On a Traditional OS (Windows/macOS): You can download the YouTube app or use the browser. The app allows offline downloads of videos for viewing without an internet connection. The interface and playback are handled locally. Similarly, you can install the Flipkart app, which loads from local storage instantly and can cache product data for browsing even when offline.

On a Cloud-Based OS (Chrome OS): You access these services exclusively through the Chrome browser. Without an internet connection, you cannot even load the homepage. Every video stream, product search, and cart update requires a round-trip to the cloud servers. All processing happens remotely, and your local device merely displays the results. If the internet fails, you lose access entirely because the apps are not running on your device.